

GX140 3in ANALOG FPV – ELECTRICAL SCHEMATIC (Rev 2.0)

METHOD: point-to-point by net label. Every conductor breaks out at the device pin.
Match net names to find the far end. No routed runs, no crossings, no junction ambiguity.

=== POWER TREE ===

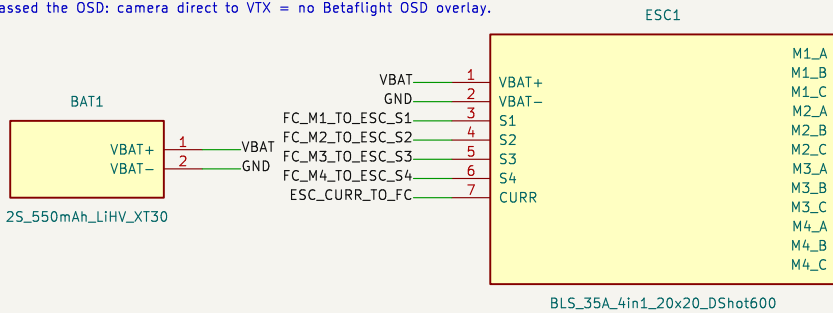
BAT XT30 -> ESC VBAT pads (22AWG). FC powered via ESC 8-pin harness (VBAT/GND/CURR/S1-S4).
FC internal 5V/3A BEC -> 5V net -> VTX, CAM, RX, GPS.

=== UART MAP (F405 Mini) ===

UART3: GPS 57600 (TX3_GPS / RX3_GPS) | UART4: SmartAudio 2.1 (TX4 only, RX4 = N/C)
UART5: ELRS CRSF 420k (TX5_CRSF -> RX.RX, RX.TX -> RX5_CRSF) | I2C: QMC5883L compass

=== VIDEO PATH (corrected) ===

CAM.VID -> FC.VIN (CAM_VIDEO) -> AT7456E OSD -> FC.VOUT -> VTX.VID_IN (VTX_VIDEO).
Grok rev1 bypassed the OSD: camera direct to VTX = no Betaflight OSD overlay.



MOTOR PHASE WIRING: 3 conductors per motor (A/B/C).
Swap any two phases to reverse direction. M1 FL CCW | M2 FR CW | M3 RL CCW | M4 RR CW.

=== STACK ASSEMBLY (bottom to top) ===

1.Bottom 2mm 2.Main 4mm 3.ESC 4.M1 4.M2x6 standoffs
5.FC F405 Mini 6.VTX Zeus(rear) 7.CAM Caddx Ant(front,TPU)
8.RX SuperD(side,tape) 9.GPS(mast,opt) 10.BAT 25(under,2x20mm strap)
Specter_1804_FL_CCW

=== TORQUE ===

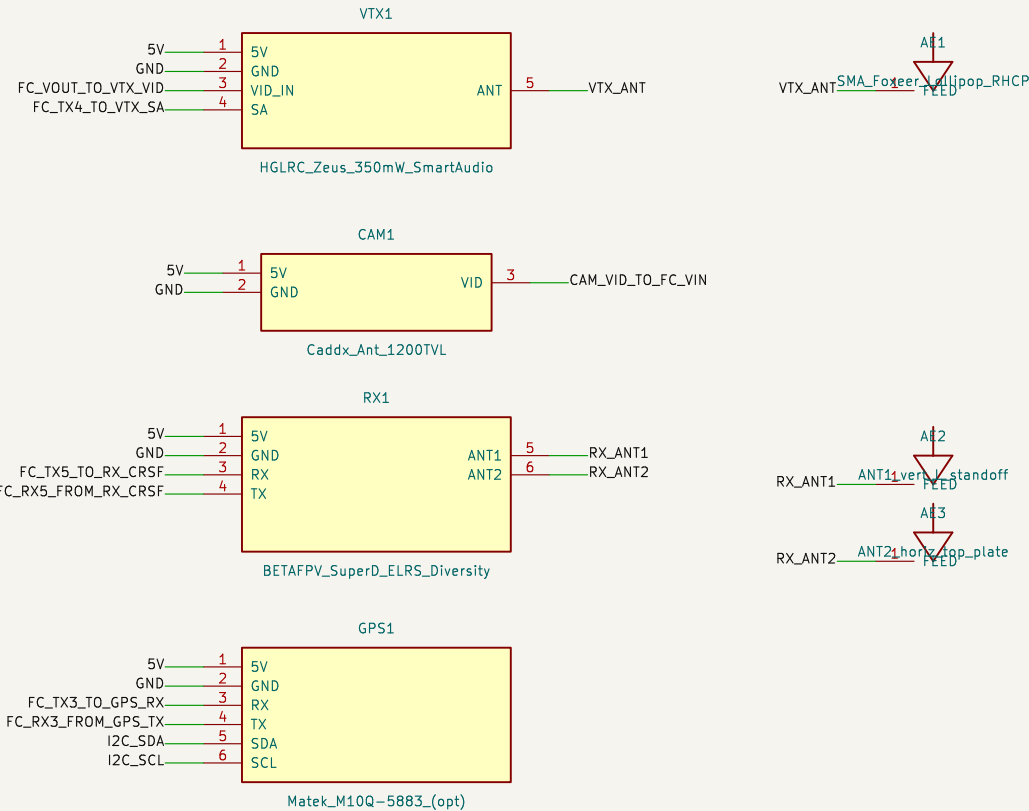
Motor M2x5: 0.25 N-m(Loctite243) Stack M2 nylon: 0.15 N-m
Top M2.5x10: 0.35 N-m Bottom M2.5x10: 0.40 N-m Prop nut: hand+1/4

=== REVERSE WIRE TAGS ===

Tag at DESTINATION end: -> DST:PIN <- SRC:COMP.PIN 5mm
1.5mm heat shrink tag, 10mm from pad, text reads toward component

=== WIRE SCHEDULE ===

PWR: BAT->ESC VBAT+/- 22AWG | BAT->FC VBAT/GND 22AWG
ARM: ESC M1-4 A/B/C -> FC VBAT/GND/CURR/S1-4 22AWG 25-55mm
HARN: ESC 8-pin -> FC VBAT/GND/CURR/S1-4 26AWG
5V: FC 5V/GND -> VTX/CAM/RX/GPS 26AWG Specter_1804_RR_CCW
VID: CAM->FC.VIN(OSD)->FC.VOUT->VTX 26AWG
UART: TX4->SA TX5->RX.RX RX5<-RX.TX TX3->GPS.RX RX3<-GPS.TX 26AWG
I2C: SDA/SCL FC->GPS compass 28AWG



=== WIRING ===

22AWG: VBAT, GND, motor phase | 26AWG: UART, DShot, video | 28AWG: I2C
JST-SH 1.0mm: CAM | JST-GH 1.25mm: GPS | XT30: battery | Heat shrink all joints

=== SAFETY ===

Smoke stopper (ShortSaver) MANDATORY first power-up | Props OFF for motor direction test
Continuity check VBAT-GND before battery | Verify motor order/direction in Betaflight Motors tab

=== BATTERY ===

2S 550mAh LiHV XT30. Under bottom plate, 2x20mm straps.

=== ANTENNAS ===

VTX: Foxeer Lollipop RHCP SMA(rear) RX: 2x dipoles 90-deg apart
RX ANT1: Vertical L-standoff(left arm) ANT2: Horizontal(top plate)

=== BETAFLIGHT ===

UART3: GPS 57600 UART4: SmartAudio auto UART5: CRSF 420k

VTX: Zeus 350mW | CAM: Caddx Ant | RX: SuperD ELRS | GPS: M10Q-5883 (opt)
Motors: HGLRC Specter 1804 3500KV | Props: GEMFAN 3052
FC: SpeedyBee F405 Mini | ESC: BLS 35A 4in1 DShot600
Frame: GX140 140mm twin 20x20, 4mm carbon | AUW ~190g
GX140 Build – net-label point-to-point method

Sheet: /
File: gx140_rev2.kicad_sch

Title: GX140 3in Analog FPV – Electrical Schematic

Size: A3 Date: 2026-06-12 Rev: 2.0
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